

Neural Plasticity in Adult Hippocampal Neurogenesis: A Systematic Review

Abstract

Adult hippocampal neurogenesis, the process by which new neurons are generated in the dentate gyrus of the adult brain, has been implicated in learning, memory consolidation, and emotional regulation. This systematic review synthesises findings from 67 peer-reviewed studies published between 2015 and 2024 examining factors that promote or inhibit neurogenesis in adult mammalian models.

1. Introduction

For decades it was believed that the adult mammalian brain was incapable of generating new neurons. This dogma was overturned in the 1990s with the discovery of active neurogenesis in the adult hippocampus and olfactory bulb. The hippocampal dentate gyrus has since become a primary focus of neuroscience research due to its role in episodic memory formation and its sensitivity to both environmental and pharmacological stimuli.

2. Promoting Factors

Physical exercise, particularly aerobic activity, consistently emerges as the most robust promoter of adult neurogenesis across species. Voluntary wheel running in rodent models increases progenitor cell proliferation by up to 200%. Environmental enrichment, caloric restriction, and intermittent fasting have also demonstrated significant pro-neurogenic effects mediated through BDNF signalling pathways and reduced cortisol exposure.

3. Inhibiting Factors

Chronic stress is the most well-documented inhibitor of adult neurogenesis, acting primarily through elevated glucocorticoid levels. Prolonged sleep deprivation reduces hippocampal progenitor cell survival by approximately 40% in murine models. Age-related decline in neurogenesis correlates with increased neuroinflammation and reduced vascular endothelial growth factor expression in the hippocampal microenvironment.

4. Clinical Implications

Deficits in adult neurogenesis have been linked to the pathophysiology of major depressive disorder and post-traumatic stress disorder. Current antidepressant treatments including SSRIs require intact hippocampal neurogenesis for their therapeutic effects, suggesting neurogenesis as a potential biomarker for treatment response. Interventions targeting neurogenesis pathways represent a promising frontier for novel psychiatric and neurodegenerative therapeutics.